

Ruxolitinib (Jakavi®)

Ruxolitinib (Jakavi[®]) is a cancer drug. It is used to treat some types of myeloproliferative neoplasms.

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What is ruxolitinib (Jakavi®)?

Ruxolitinib is also called Jakavi[®]. It is a type of targeted therapy drug called a Janus Associated Kinase, or JAK inhibitor.

Ruxolitinib is used to treat [myelofibrosis](#) and [polycythaemia vera \(PV\)](#). It may sometimes be used to treat graft versus host disease (GvHD) after having a [stem cell transplant](#).

It is best to read this information with our general information about [targeted therapy](#) drugs and [blood cancers](#).

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How ruxolitinib is given

Ruxolitinib comes in tablets. This means you can take it at home.

During a course of treatment, you will meet someone from your cancer team, such as a:

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- blood cancer doctor (haematologist)
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your targeted therapy.

Your course of treatment

The nurse or pharmacist will give you the ruxolitinib tablets to take home. They may give you tablets of different strengths. Always take them exactly as explained. This is important to make sure they work as well as possible for you.

Your nurse or pharmacist may also give you other drugs to take home. Take all your drugs and medicines exactly as they have been explained to you.

Your doctor may need to change the dose of ruxolitinib throughout your treatment. This depends on the results of your blood tests.

You usually continue taking ruxolitinib for as long as it is working well for you and any side effects can be managed. It is important that you do not stop taking it without talking to your doctor first. Stopping the drug suddenly may make you feel unwell. Doctors usually reduce your dose before stopping it completely.

If you are having ruxolitinib for GvHD, your doctor may reduce your dose over a few months.

Taking ruxolitinib tablets

You usually take ruxolitinib tablets twice a day. You can take the tablets with or without food.

Make sure you:

- take them at the same time every day
- swallow them whole with a glass of water
- do not chew, break or crush them

Speak to your doctor or pharmacist if you find it difficult to swallow the tablets.

If you forget to take the tablets, take the next dose at the usual time. Do not take a double dose to make up for the missed dose.

Other things to remember about your tablets:

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
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If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.

- Do not throw away unused tablets. Return them to your cancer team at the hospital.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

You will have regular blood tests during treatment. If your white blood cell count is low, your doctor may stop your treatment for a short time, until your cell count increases. Sometimes they will reduce the dose of your treatment.

Shingles

Some people can get shingles (herpes zoster) while taking ruxolitinib. This is an infection of a nerve and the skin around it. It is caused by the same virus that causes chickenpox.

Tell your doctor straight away if you develop any of the following on your skin:

- a burning feeling
- tingling
- numbness
- itchiness
- a rash with blisters.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. Sometimes this can be serious. If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. This can include:

- nosebleeds
- bleeding gums

- heavy periods
- vomit blood
- vomit that looks like coffee grounds
- pass black or bloodstained stools (poo)
- blood in your urine (pee)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Dizziness

This treatment can cause dizziness. Tell your doctor or nurse if this happens. It is important that you do not drive or operate machinery if you feel dizzy.

Raised cholesterol level

This treatment can raise the level of cholesterol. This is a fatty substance in the blood. You may have a [blood test](#) to check the level of cholesterol. Your doctor may refer you to a dietitian for advice on diet, or prescribe medicines to control your cholesterol levels.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Feeling sick

You may feel sick with this treatment. This may be more likely if you are having it for GvHD. You may be given anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Increased blood pressure

Ruxolitinib can cause high blood pressure. Your blood pressure will be checked regularly. Tell your doctor if you already have high blood pressure and are on treatment to control it.

Changes in the way your liver works

Ruxolitinib can affect how your liver works. You will have regular [blood tests](#) during treatment to check how well your liver is working. Some people may be given a lower dose of ruxolitinib if they have or have had liver problems in the past.

Weight gain

Ruxolitinib may cause weight gain. Talk to your doctor if this happens.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Risk of developing skin cancer (non-melanoma)

A small number of people taking ruxolitinib develop skin cancer. Doctors are not sure if this is directly linked to the drug. There may be other reasons. Your doctor will discuss the risk of developing [skin cancer](#) with you. Tell your doctor if you notice any changes in your skin. They may recommend taking extra care in the sun.

Effects on the brain

Sometimes this drug might cause a bleed or a serious infection in the brain. Contact the hospital on the 24-hour number you have been given straight away or call 999 if you or someone else notices you have:

- a severe headache

- weakness in your body
- changes to your speech
- feeling drowsy or confused
- changes to your vision
- have difficulty with your balance or walking.

Excess wind

Some people pass more wind than usual when taking ruxolitinib. If you are affected, try to avoid food that gives you wind, for example:

- beans
- pickles
- fizzy drinks.

Hepatitis reactivation

Tell your doctor if you have had Hepatitis B or C (a liver infection) in the past. Rarely this treatment can make it active again. Your doctor or nurse will talk to you about this and may test you for it.

Tuberculosis (TB)

Tell your doctor if you have had tuberculosis (TB) in the past. Rarely this treatment can make it active again. You should also tell your doctor if you been in close contact with someone who has TB.

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
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- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help